

The ethics of artificial intelligence

Same tool. Three different ethical problems. Keep Priya's team in mind — every concept below maps onto something that happened to MuseBot.

What this key knowledge covers

This summary distils everything you need for **KK13 — The ethics of artificial intelligence** in Innovative Solutions. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

What we actually mean by "ethics of AI"

Ordinary software is **deterministic**: a programmer writes the rules, so if it behaves badly you can usually point to the line that did it. Generative AI is different. It is *trained*, not hand-coded — it absorbs statistical patterns from enormous datasets and then produces fresh text, images, audio or code that no one wrote directly.

That creates a gap between **what the makers intended** and **what the system actually does**. *Ethics of AI* is the study of the moral questions that live in that gap: who could be harmed, who is responsible, and what we owe the people whose data trained the model and the people who use its output.

VCAA narrows "ethics of AI" to **three examinable issues**. Memorise them as a set — most questions ask you to pick one and reason about it for a specific scenario:

AI output that is discriminatory, unsafe, false or distressing — because of what it learned or how it's used.

AI that lowers the skill needed to write phishing, malware, scams or deepfakes — arming people who couldn't before.

AI trained on protected works that then reproduces or imitates them, raising ownership and consent questions.

Think of a generative model as an apprentice who learned by copying millions of examples in a workshop. They become brilliantly skilled — but they also pick up the workshop's **bad habits** (bias), they'll happily forge a document if you ask the right way (cyber misuse), and their "original" work is stitched from everything they ever studied, including pieces they had no permission to copy (copyright). The apprentice isn't evil. The ethics live in how they were trained and how we let them be used.

Biased or harmful content

A model only knows the world its training data showed it. If that data over-represents one group, era or viewpoint, the model treats that as "normal" — and quietly bakes the imbalance into everything it makes. This is *algorithmic bias*: unfair, systematic skew in a system's outputs that traces back to its data or design, not to a single typo.

Two flavours show up in the exam:

Bias doesn't just pass through a model — it can get **amplified**. If 60% of "doctor" photos in the data show one group, the model often outputs that group far more than 60% of the time, because it learns to play the safest bet. Mildly skewed data in, strongly skewed world out. Try it below.

Why it matters ethically

Biased and harmful output isn't just embarrassing — it can entrench discrimination at scale (a screening tool that quietly down-ranks certain names), spread unsafe information faster than any human could, or destroy a real person's reputation through a deepfake. Because AI runs at scale, a single skewed pattern can touch millions of decisions. That's why Australia's *AI Ethics Principles* put **fairness, reliability & safety**, and **human wellbeing** right at the front.

Content usable for cyber attacks

Generative AI is a brilliant writer, coder and impersonator. Those same talents make it a force-multiplier for attackers. The core ethical concern is **lowered barriers**: tasks that once needed real skill — fluent phishing in perfect English, working malware, a cloned voice — are now a well-phrased request away.

Flawless, personalised scam emails and texts — like MuseBot's fake "Principal" message — at massive volume.

Draft malware, exploit scripts or password-cracking helpers for someone who couldn't write them alone.

Impersonate a real person in audio or video to authorise a transfer or trick a victim.

Responsible AI products fit *guardrails* — filters and refusals that block obvious misuse. The catch is the *jailbreak* (or *prompt injection*): wrapping a banned request in a story, role-play or "for a security class" framing so the filter waves it through. The MuseBot team learned this the hard way when a student dressed up "write a phishing email" as "write an example email for our cyber-safety poster."

Capability (the AI *can* write a phishing email) is not the same as a **cyber attack** (someone *sends* it to steal logins). In an answer, name the AI capability *and* the attack it enables — "MuseBot can generate convincing, error-free phishing emails, which lets an attacker run a credential-harvesting scam against staff at a scale they couldn't manage by hand."

Generating content from copyright material

To learn, generative models are fed staggering amounts of human work — books, art, articles, music, code — much of it protected by copyright and used **without the creator's permission or payment**. The output can then reproduce a source almost verbatim, or imitate a living artist's style so closely it competes with them. Three ethical knots:

Copyright is the creator's automatic legal right to control copying of their original work (in Australia, the **Copyright Act 1968**). A *derivative work* builds on an existing work. *Plagiarism* is passing off someone's work as your own — an *ethical/academic* breach that can happen even when no copyright is broken. A strong answer keeps "illegal" and "unethical" separate: AI imitation can be unethical and harmful to creators even where the legal position is murky.

So who is actually responsible?

"The AI did it" is never a complete answer — and examiners pounce on students who stop there. Responsibility is shared along a chain, and a top-band answer names where it sits:

When a question says "evaluate" or "discuss," brace your judgement against named principles — it instantly lifts your response. Australia's 8 voluntary **AI Ethics Principles** include:

Fairness (no unfair discrimination) · **Reliability & safety** (works as intended without harm) · **Transparency & explainability** (people know AI is involved and roughly how) · **Privacy protection & security** · **Contestability** (you can challenge an AI decision) · **Accountability** (a human is answerable) · **Human-centred values** · **Human, societal & environmental wellbeing**. For KK13, **fairness, safety, transparency** and **accountability** do most of the work.

MuseBot must not discriminate — it should produce equitable results across genders, backgrounds and accents.

It must work as intended without causing harm — no unsafe advice, no dangerous output.

Users should know they're dealing with AI and roughly how it reached an output.

Priya's team and the school stay answerable for what MuseBot does — responsibility can't be offloaded to 'the algorithm'.

A student affected by an unfair AI decision must be able to challenge it and reach a human.

Key terms — tap to flip

These are the words that turn a vague answer into a precise one. If you can deploy them correctly, you're writing at a band-5+ level.

A systematic, unfair skew in an AI system's outputs that traces back to its training data or design — not a one-off error.

The large body of examples a model learns patterns from. Its gaps and imbalances become the model's blind spots.

AI that produces new content — text, images, audio, code — by predicting likely patterns, rather than retrieving a fixed answer.

When a model states something false or invented with total confidence, because it predicts plausible-sounding text, not verified fact.

Synthetic media — image, audio or video — that convincingly impersonates a real person, often used to deceive or harm.

A built-in filter or refusal that stops a model producing disallowed content, such as malware or hate speech.

A prompt crafted to slip past guardrails — disguising a banned request as fiction, role-play or 'an example'. Also called prompt injection.

Read the command word, then aim your answer

Half of all lost marks in this topic come from answering the wrong question — describing when you were asked to evaluate, or listing when you were asked to justify. Decode the verb first.

Name it. A word or short phrase — no explanation needed.

Say what it is and add detail or features. Still no judgement.

Show how or why — give the cause-and-effect, not just the what.

Weigh more than one side or angle; show the tensions before any conclusion.

Make a judgement and back it with reasons — strengths vs limitations, then a verdict.

Give the reasons a choice is the right one; defend it against the alternative.

Propose a course of action and say why it suits this specific situation.

The P·E·L writing trainer

Most ethics answers are graded on structure as much as content. Use **P·E·L**: make a *Point* (name the issue), give *Evidence* (tie it to the scenario), then *Link* back to impact, a principle or a who's-responsible judgement. Write a response below — the trainer checks for all three moves.

VCAA-style questions

Attempt each in your head or on paper first. Then reveal the weak answer (to see the trap), the strong answer (to see the standard), and the marking guide. Self-mark honestly — your score builds in the dock.

If you remember six things

- **Three issues:** biased/harmful content · content usable for cyber attacks · generating from copyright material.
- **Bias comes from data and gets amplified** — the model can output more skew than it was fed.
- **Cyber risk = lowered barriers:** name the AI capability *and* the attack it enables; guardrails help but jailbreaks bypass them.
- **Copyright:** separate consent, compensation and ownership — and keep "illegal" apart from "unethical."
- **"The AI did it" is never enough** — pin responsibility on developers, makers, users or deployers.
- **Anchor judgements** to the AI Ethics Principles: fairness, safety, transparency, accountability.

Digital Vector · VCE Applied Computing Units 1 & 2 · U2·O1 Innovative Solutions · KK13 Ethics of AI. Built as a single offline file. **MuseBot, Brookvale Secondary and all named students are fictional** teaching scenarios and can be renamed. Aligned to the VCAA Applied Computing Study Design 2025–2028; this is a teaching resource, not a VCAA publication.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a

judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Define

What is meant by **algorithmic bias** and give one example of how it could arise in MuseBot.

✗ A weak answer

tap to open “Algorithmic bias is when the AI is biased and unfair to people. For example MuseBot is sometimes wrong about things.” No real definition (circular), and the example isn't bias — it's just 'wrong'. 0–1 marks.

✓ A strong answer

tap to open **Definition:** Algorithmic bias is a systematic, unfair skew in an AI system's outputs that comes from imbalances in its training data or design. **Example:** If MuseBot's image data shows mostly one gender as “leaders,” it will keep drawing school captains as that gender — an unfair pattern produced by the data, not a random error.

How the marks are awarded

- 1 mark — a correct definition referencing systematic/unfair skew from data or design.
- 1 mark — a relevant MuseBot example that is genuinely bias (a skewed pattern), not just an error or a different issue.

Q2. Explain

During launch week, a student used MuseBot to produce an email impersonating the Principal that asked staff to “confirm your login details.” Explain how a generative AI tool like MuseBot could produce content that is **usable for a cyber attack**.

✗ A weak answer

tap to open “MuseBot is dangerous because hackers could use it. AI is bad for security and could be used to hack the school. People should be careful with AI.” Vague and unsupported. Names no capability, no attack type, no mechanism. 1 mark at most.

✓ A strong answer

tap to open MuseBot can generate fluent, error-free, personalised text on demand — a capability attackers exploit. This lets someone produce a convincing phishing email impersonating the Principal far faster and more believably than they could alone, enabling a credential-harvesting attack against staff. Because the AI lowers the skill and time needed, attacks can be run at scale, and guardrails can be bypassed with a jailbreak (e.g. framing it as “an example for a poster”). The same ability extends to drafting malicious scripts or cloning a voice for impersonation.

How the marks are awarded

- 1 mark — identifies a relevant AI capability (e.g. generating fluent/personalised text or code).
- 1 mark — links it to a specific attack (phishing / malware / impersonation).
- 1 mark — explains the mechanism of harm: lowered barrier, scale, or believability.
- 1 mark — depth: jailbreak/guardrail bypass, or a second attack vector.

Q3. Discuss

A poster MuseBot generated for the school fair is unmistakably in the style of a living, working illustrator. **Discuss** the copyright and ethical issues raised when MuseBot generates content in the style of an existing creator.

✗ A weak answer

tap to open “It's copying so it's illegal and that's stealing. The team should just not do that because it's against copyright law.” One-sided and over-confident on the law. No discussion of consent, compensation, ownership, or the legal/ethical distinction. 1-2 marks.

✓ A strong answer

tap to open MuseBot learns from protected works, so its output can imitate a creator's style closely enough to compete with them. **Consent:** the illustrator likely never agreed to their work training the model — an ethical breach even where copying a style isn't clearly illegal. **Compensation & fairness:** the creator earns nothing while the tool profits from their visual language, breaching the fairness principle. **Ownership is unsettled:** it's unclear whether the student, the team or the original artists hold rights to the output — so a careful answer separates what is unethical from what is definitely illegal.

How the marks are awarded

- 1 mark — recognises training on / imitation of protected creative work.
- 1 mark — raises consent OR compensation for the original creator.
- 1 mark — addresses ownership/rights uncertainty of AI output.
- 1 mark — discusses more than one angle and distinguishes legal vs ethical.

Q4. Evaluate

A Year 9 student used MuseBot's "health tips" feature and received advice that, if followed, could be physically harmful. **Evaluate** the ethical responsibilities of Priya's team in this situation. Refer to at least **two** of Australia's AI Ethics Principles.

✗ A weak answer

tap to open "The team should say sorry and take it down. It's not really their fault because the AI made the advice, not them, and the student chose to use it." Offloads blame to 'the AI' and the user — the opposite of the expected judgement. Names no principles. 1 mark.

✓ A strong answer

tap to open The team chose to release MuseBot to students, so under **accountability** they — not "the AI" — are answerable for foreseeable harm. Shipping a health feature without safety filtering breaches **reliability & safety**: the tool caused harm rather than working safely as intended. There is also a **transparency** failure if students weren't warned the advice was AI-generated and possibly wrong. **Judgement**: their responsibility is high — they had a duty to test for harmful output, add guardrails and disclaimers, and provide a human to contest results; the student's choice reduces but does not remove that duty. A balanced answer notes the model maker shares responsibility for the underlying training, but the deployers remain primarily accountable.

How the marks are awarded

- 1 mark — identifies the team's responsibility as developers/deployers.
- 1 mark each — correctly applies TWO named AI Ethics Principles to the scenario (max 2).
- 1 mark — reaches a clear, reasoned judgement about the level of responsibility.
- 1 mark — balance: acknowledges shared responsibility (model maker / user) without excusing the team.

Revision checklist

- Three issues: biased/harmful content · content usable for cyber attacks · generating from copyright material.
- Bias comes from data and gets amplified — the model can output more skew than it was fed.
- Cyber risk = lowered barriers: name the AI capability and the attack it enables; guardrails help but jailbreaks bypass them.
- Copyright: separate consent, compensation and ownership — and keep "illegal" apart from "unethical."
- "The AI did it" is never enough — pin responsibility on developers, makers, users or deployers.
- Anchor judgements to the AI Ethics Principles: fairness, safety, transparency, accountability.